
Linear Statistical Models

B. Statistical Data Science 2nd Year Indian Statistical Institute

Teacher: Soham Sarkar

Exercise Series 8

The following was covered in the class. This is given for your practice. You may skip it for now.

Exercise 1. Consider the one-way random effects model: $y_{ij} = \mu + t_i + \epsilon_{ij}$, $i = 1, \dots, I$, $j = 1, \dots, J$, where $t_i \stackrel{\text{i.i.d.}}{\sim} \mathbf{N}(0, \sigma_T^2)$ and $\epsilon_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathbf{N}(0, \sigma_e^2)$ are independent random variables.

(a) Find $\text{Var}(y_{ij})$ and $\text{Cov}(y_{ij}, y_{i'j'})$.

We defined the following sum-of-squares as in the one-way fixed effects model

$$\text{SS}_{\text{tr}} = J \sum_{i=1}^I (\bar{y}_{i.} - \bar{y}_{..})^2, \quad \text{SSE} = \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \bar{y}_{i.})^2, \quad \text{and} \quad \text{TSS} = \text{SS}_{\text{tr}} + \text{SSE} = \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \bar{y}_{..})^2.$$

(b) Find the expected values of SS_{tr} , SSE and TSS .

(c) Show that SS_{tr} and SSE are independent, and they follow central χ^2 distributions with different scales (i.e., the distributions are of the form $\sigma^2 \chi_k^2$ for some σ^2 and k). Also, determine the degrees of freedom (k) of the distributions in each case.

(d) Find the distribution of $T = \text{SS}_{\text{tr}}/\text{SSE}$. From this, deduce that a right-tailed test based on T is appropriate for testing $\mathcal{H}_0 : \sigma_T^2 = 0$ against $\mathcal{H}_a : \sigma_T^2 > 0$. Find a suitable cut-off that ensures that the test is of level $\alpha \in (0, 1)$. Show that this test is unbiased.

In the following, we study a special type of two-way ANOVA (fixed effects) model. The results developed here will be used subsequently.

Exercise 2. Consider the following two-way ANOVA model with one observation per cell:

$$y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}, \quad i = 1, \dots, I, \quad j = 1, \dots, J.$$

Also assume that $\epsilon_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathbf{N}(0, \sigma^2)$.

(a) Show that $\text{SSE} = \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..})^2$ and it has degrees of freedom $(I-1)(J-1)$.

(b) Use the results proved in the class to show that $\text{SSE} \sim \sigma^2 \chi_{(I-1)(J-1)}^2$.

(c) Verify that $\text{SSE} = \sum_{i=1}^I \sum_{j=1}^J (\epsilon_{ij} - \bar{\epsilon}_{i.} - \bar{\epsilon}_{.j} + \bar{\epsilon}_{..})^2$.

(d) Conclude that if $Z_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathbf{N}(0, \sigma^2)$, then $\sum_{i=1}^I \sum_{j=1}^J (Z_{ij} - \bar{Z}_{i.} - \bar{Z}_{.j} + \bar{Z}_{..})^2 \sim \sigma^2 \chi_{(I-1)(J-1)}^2$.

(e) Use the form of SSE in a two-way ANOVA model with interaction to show that if $Z_{ijk} \stackrel{\text{i.i.d.}}{\sim} \mathbf{N}(0, \sigma^2)$, $i = 1, \dots, I$, $j = 1, \dots, J$, $k = 1, \dots, K$, then $\sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (Z_{ijk} - \bar{Z}_{ij.})^2 \sim \sigma^2 \chi_{I(J(K-1))}^2$.

Note: Alternatively, notice that $\xi_{ij} = \sum_{k=1}^K (Z_{ijk} - \bar{Z}_{ij.})^2$, $i = 1, \dots, I$, $j = 1, \dots, J$ are independent. Also, for each (i, j) , $\xi_{ij} \sim \sigma^2 \chi_{K-1}^2$.

Exercise 3. Consider the following (balanced) two-way random effects model:

$$y_{ijk} = \mu + a_i + b_j + c_{ij} + \epsilon_{ijk}, i = 1, \dots, I, j = 1, \dots, J, k = 1, \dots, K,$$

where $a_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_A^2)$, $b_j \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_B^2)$, $c_{ij} \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_{AB}^2)$ and $\epsilon_{ijk} \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_e^2)$ are independent.

(a) Calculate $\text{Var}(y_{ijk})$ and $\text{Cov}(y_{ijk}, y_{i'j'k'})$.

The above model was defined in the class, and we discussed the different sums of squares (SSs) associated with the model.

$$\begin{aligned} \text{SSA} &= JK \sum_{i=1}^I (\bar{y}_{i..} - \bar{y}_{...})^2, & \text{SSB} &= IK \sum_{j=1}^J (\bar{y}_{.j.} - \bar{y}_{...})^2, \\ \text{SSAB} &= K \sum_{i=1}^I \sum_{j=1}^J (\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...})^2, & \text{SSE} &= \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (y_{ijk} - \bar{y}_{ij.})^2. \end{aligned}$$

(b) Show that

$$\begin{aligned} \text{SSA} &= JK \sum_{i=1}^I (f_i - \bar{f})^2, & \text{SSB} &= IK \sum_{j=1}^J (g_j - \bar{g})^2 \\ \text{SSAB} &= K \sum_{i=1}^I \sum_{j=1}^J (h_{ij} - \bar{h}_{i.} - \bar{h}_{.j} + \bar{h}_{..})^2, & \text{SSE} &= \sum_{i=1}^I \sum_{j=1}^J \sum_{k=1}^K (\epsilon_{ijk} - \bar{\epsilon}_{ij.})^2, \end{aligned}$$

where $f_i = a_i + \bar{c}_{i.} + \bar{\epsilon}_{i..}$, $g_j = b_j + \bar{c}_{.j} + \bar{\epsilon}_{.j.}$, and $h_{ij} = c_{ij} + \bar{\epsilon}_{ij.}$.

(c) Verify that $f_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_F^2)$, where $\sigma_F^2 = \sigma_A^2 + \frac{\sigma_{AB}^2}{J} + \frac{\sigma_e^2}{JK}$. Similarly, $g_j \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_G^2)$, where $\sigma_G^2 = \sigma_B^2 + \frac{\sigma_{AB}^2}{I} + \frac{\sigma_e^2}{IK}$. Also, $h_{ij} \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma_H^2)$, where $\sigma_H^2 = \sigma_{AB}^2 + \frac{\sigma_e^2}{K}$.

(d) Use parts (a) and (b) to show that $\text{SSA} \sim JK\sigma_F^2\chi_{I-1}^2$ and $\text{SSB} \sim IK\sigma_G^2\chi_{J-1}^2$.

(e) Use Exercise 1, part (d) to show that $\text{SSAB} \sim K\sigma_H^2\chi_{(I-1)(J-1)}^2$.

(f) Use Exercise 1, part (e) to show that $\text{SSE} \sim \sigma_e^2\chi_{I(J(K-1))}^2$.

(g) Show that

$$\begin{aligned} \text{Cov}(f_i - \bar{f}, g_j - \bar{g}) &= \text{Cov}(\bar{c}_{i.} - \bar{c}_{..}, \bar{c}_{.j} - \bar{c}_{..}) + \text{Cov}(\bar{\epsilon}_{i..} - \bar{\epsilon}_{...}, \bar{\epsilon}_{.j.} - \bar{\epsilon}_{...}), \\ \text{Cov}(f_i - \bar{f}, h_{ij} - \bar{h}_{i.} - \bar{h}_{.j} + \bar{h}_{..}) &= \text{Cov}(\bar{c}_{i.} - \bar{c}_{..}, c_{ij} - \bar{c}_{i.} - \bar{c}_{.j} + \bar{c}_{..}) \\ &\quad + \text{Cov}(\bar{\epsilon}_{i..} - \bar{\epsilon}_{...}, \bar{\epsilon}_{ij.} - \bar{\epsilon}_{i..} - \bar{\epsilon}_{.j.} + \bar{\epsilon}_{...}), \\ \text{Cov}(g_j - \bar{g}, h_{ij} - \bar{h}_{i.} - \bar{h}_{.j} + \bar{h}_{..}) &= \text{Cov}(\bar{c}_{.j} - \bar{c}_{..}, c_{ij} - \bar{c}_{i.} - \bar{c}_{.j} + \bar{c}_{..}) \\ &\quad + \text{Cov}(\bar{\epsilon}_{.j.} - \bar{\epsilon}_{...}, \bar{\epsilon}_{ij.} - \bar{\epsilon}_{i..} - \bar{\epsilon}_{.j.} + \bar{\epsilon}_{...}), \\ \text{Cov}(f_i - \bar{f}, \epsilon_{ijk} - \bar{\epsilon}_{ij.}) &= \text{Cov}(\bar{\epsilon}_{i..} - \bar{\epsilon}_{...}, \epsilon_{ijk} - \bar{\epsilon}_{ij.}) \\ \text{Cov}(g_j - \bar{g}, \epsilon_{ijk} - \bar{\epsilon}_{ij.}) &= \text{Cov}(\bar{\epsilon}_{.j.} - \bar{\epsilon}_{...}, \epsilon_{ijk} - \bar{\epsilon}_{ij.}) \\ \text{Cov}(h_{ij} - \bar{h}_{i.} - \bar{h}_{.j} + \bar{h}_{..}, \epsilon_{ijk} - \bar{\epsilon}_{ij.}) &= \text{Cov}(\bar{\epsilon}_{ij.} - \bar{\epsilon}_{i..} - \bar{\epsilon}_{.j.} + \bar{\epsilon}_{...}, \epsilon_{ijk} - \bar{\epsilon}_{ij.}). \end{aligned}$$

Hint: All the random variables a_i , b_j , c_{ij} , ϵ_{ijk} are independent.

(h) Show that all the covariance terms on the right side in the previous step are 0.

Hint: You can use direct calculations. Otherwise, notice that $\text{Cov}(c_{ij} - \bar{c}_{i.}, \bar{c}_{i.} - \bar{c}_{..}) = 0$. This can be used to show that $\text{Cov}(\bar{c}_{i.} - \bar{c}_{..}, \bar{c}_{.j} - \bar{c}_{..}) = 0$. The other terms can be simplified similarly.

(i) Use the previous part to conclude that SSA, SSB, SSAB and SSE are independent.

(j) Using the previous part and parts (d)–(f), show that

$$\begin{aligned}
T_A &= \frac{\text{MSA}}{\text{MSAB}} := \frac{\text{SSA}/(I-1)}{\text{SSAB}/((I-1)(J-1))} \sim (1 + \nu_A)F_{I-1, (I-1)(J-1)}, \quad \nu_A \geq 0, \text{ “=” iff } \sigma_A^2 = 0, \\
T_B &= \frac{\text{MSB}}{\text{MSAB}} := \frac{\text{SSB}/(J-1)}{\text{SSAB}/((I-1)(J-1))} \sim (1 + \nu_B)F_{J-1, (I-1)(J-1)}, \quad \nu_B \geq 0, \text{ “=” iff } \sigma_B^2 = 0, \\
T_{AB} &= \frac{\text{MSAB}}{\text{MSE}} := \frac{\text{SSAB}/(I-1)(J-1)}{\text{SSE}/IJ(K-1)} \sim (1 + \nu_{AB})F_{(I-1)(J-1), IJ(K-1)}, \\
&\hspace{25em} \nu_{AB} \geq 0, \text{ “=” iff } \sigma_{AB}^2 = 0, \\
\tilde{T}_A &= \frac{\text{MSA}}{\text{MSE}} \sim (1 + \tilde{\nu}_A)F_{I-1, IJ(K-1)}, \quad \tilde{\nu}_A \geq 0, \text{ “=” iff } \sigma_A^2 = 0 \text{ and } \sigma_{AB}^2 = 0, \\
\tilde{T}_B &= \frac{\text{MSB}}{\text{MSE}} \sim (1 + \tilde{\nu}_B)F_{J-1, IJ(K-1)}, \quad \tilde{\nu}_B \geq 0, \text{ “=” iff } \sigma_B^2 = 0 \text{ and } \sigma_{AB}^2 = 0.
\end{aligned}$$

(k) Based on the conclusion in the previous part, comment on the appropriateness of using $T_A/\tilde{T}_A$ for testing $\mathcal{H}_0^{(A)} : \sigma_A^2 = 0$ vs. $\mathcal{H}_a^{(A)} : \sigma_A^2 > 0$. Similarly, comment on the appropriateness of using $T_B/\tilde{T}_B$ for testing $\mathcal{H}_0^{(B)} : \sigma_B^2 = 0$ vs. $\mathcal{H}_a^{(B)} : \sigma_B^2 > 0$. How would you perform these tests? Construct a test for $\mathcal{H}_0^{(AB)} : \sigma_{AB}^2 = 0$ vs. $\mathcal{H}_a^{(AB)} : \sigma_{AB}^2 > 0$.

(l) Use parts (d)–(f) to show that

$$\mathbb{E}(\text{MSA}) = JK\sigma_F^2, \quad \mathbb{E}(\text{MSB}) = IK\sigma_G^2, \quad \mathbb{E}(\text{MSAB}) = K\sigma_H^2, \quad \mathbb{E}(\text{MSE}) = \sigma_e^2.$$

Use these to find unbiased estimators of the *variance components* σ_A^2 , σ_B^2 , σ_{AB}^2 and σ_e^2 .

Exercise 4 (Practical 4). Consider the `pima` dataset, which can also be found in the R package `faraway`. We are interested in modeling the `bmi` of female Pima Indians depending on the number of pregnancies `pregnant`.

(a) Do you see any anomaly in the data? Use `summary` to get a summary of the data. Can `bmi` be zero?

Note: Missing or anomalous data is not always clearly declared in a dataset. You should always check the values for possible errors and/or issues before analyzing the data.

(b) Treat `pregnant` as a factor (categorical), but consider binning the variable into 6 categories: 0, 1, 2, 3-4, 5-6, and 7 or more pregnancies.

(c) Consider the other variables and decide whether they have significant impact on the relationship between `bmi` and `pregnant`.

(d) Based on your model, check whether `bmi` depends on `pregnant` or not.

Exercise 5 (Practical 5). The `lgp-survey` data set contains measurements on the height of several students along with the heights of their fathers and mothers. It also contains a variable called `HasY`, which indicates whether the student has the Y-chromosome (thus indicating the biological gender). We are interested in finding the relationship between the height of a student and the heights of her/his father and mother.

(a) Fit a linear model with the height of the students as response and the heights of her/his father and mother as covariates. What do you see? Which covariates are significant? Do the results seem reasonable to you?

- (b) Let us now take the gender of the students into account. Add `HasY` as a covariate in your model and fit it using `lm`. Can you understand what the fitted model looks like? Does this new model seem more reasonable?
- (c) In the previous part, our model only differs for the two genders in terms of the intercept. In order to have two different models for the two genders, you can consider a model with *interactions* between heights of father and mother and the gender of the student. You can use

```
lm(height ~ height_father + height_mother + HasY
    + height_father:HasY + height_mother:HasY)
```

to fit the desired model. What do the parameters represent? Do the estimates seem more reasonable now?

- (d) Check the data. Do you see anything that may cause a problem? What about the number of participants in the study?
- (e) Based on the data, test the following hypothesis:

The height of a male student does not depend on the height of his mother and the height of a female student does not depend on the height of her father.

Exercise 6 (Will be covered in the next class). Consider the analysis of covariance (ANOCOVA) model $\mathbf{y} = \mathbf{Z}\boldsymbol{\alpha} + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, where $\mathbf{Z}$ is a matrix of 0's and 1's (corresponding to the intercept and the factors) and $\mathbf{X}$ is the matrix containing the numerical covariates. The normal equations in this case turn out to be

$$\begin{aligned} \mathbf{Z}^\top \mathbf{Z} \boldsymbol{\alpha} + \mathbf{Z}^\top \mathbf{X} \boldsymbol{\beta} &= \mathbf{Z}^\top \mathbf{y} \\ \mathbf{X}^\top \mathbf{Z} \boldsymbol{\alpha} + \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} &= \mathbf{X}^\top \mathbf{y}. \end{aligned}$$

- (a) Show that if $\mathcal{C}(\mathbf{Z})$ and $\mathcal{C}(\mathbf{X})$ are linearly independent, then $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top (\mathbf{I} - \mathbf{P}_Z) \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{I} - \mathbf{P}_Z) \mathbf{y}$ and $\hat{\boldsymbol{\alpha}} = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}})$ are solutions to the normal equations above.

Note: The condition of linear independence of $\mathcal{C}(\mathbf{Z})$ and $\mathcal{C}(\mathbf{X})$ is needed for $\mathbf{X}^\top (\mathbf{I} - \mathbf{P}_Z) \mathbf{X}$ to be non-singular.

- (b) Compare these solutions to the solutions obtained from individual ANOVA and regression models:

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\alpha} + \boldsymbol{\epsilon}_1 \quad \text{and} \quad \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}_2.$$

- (c) Show that the sum of squared errors (SSE) of the ANOCOVA model can be written as

$$\text{SSE} = \text{SSE}_Z - \mathbf{e}_{xy}^\top \mathbf{E}_{xx}^{-1} \mathbf{e}_{xy},$$

where SSE_Z is the SSE of the ANOVA model $\mathbf{y} = \mathbf{Z}\boldsymbol{\alpha} + \boldsymbol{\epsilon}_1$, $\mathbf{E}_{xx} = \mathbf{X}^\top (\mathbf{I} - \mathbf{P}_Z) \mathbf{X}$ and $\mathbf{e}_{xy} = \mathbf{X}^\top (\mathbf{I} - \mathbf{P}_Z) \mathbf{y}$.

Hint: The SSE of a linear model $\mathbf{y} = \mathbf{G}\boldsymbol{\theta} + \boldsymbol{\eta}$ is $\mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\theta}}^\top \mathbf{G}^\top \mathbf{G} \hat{\boldsymbol{\theta}}$, where $\hat{\boldsymbol{\theta}}$ is a solution to the normal equations. From the normal equations, we also have $\hat{\boldsymbol{\theta}}^\top \mathbf{G}^\top \mathbf{G} \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}^\top \mathbf{G}^\top \mathbf{y}$. So, the SSE can be alternatively expressed as $\mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\theta}}^\top \mathbf{G}^\top \mathbf{y}$.